

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019- 2020 (Odd Semester)

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Degree, Semester & Branch: III Semester B.E. Computer Science and Engineering

Course Code & Title: CS8391 Data Structures

Name of the Faculty member: Mrs.B.Vijayalakshmi, AP(SG)/CSE

Innovative practice: Active Learning- Crossword Puzzle

Topic: Linear Data Structures- List

Objectives of the activity:

- To enable the students to familiarize the concepts of linear data structure and improve the problem solving skills.

Justification for choosing the topic:

Instead of carrying out regular multiple choice question test to identify the understanding level of a student, a game based activity was chosen to do the revision of unit I.

Activity:

A crossword is an interesting word puzzle activity which helps the students to recollect the concept and try to find out the answer individually. It is one type of game, so this an enjoyable activity rather than regular multiple choice questions test.

The students were given with a crossword puzzle of 10 questions which was displayed to them individually. They have to think and provide the answers in a paper. The activity was conducted in CC IV for duration of 15 minutes as shown in figure 1 &2.

In the Unit I- revision hour the topics in the unit were revised with the help of this crossword puzzle. The students answered the questions in the starting 15 minutes of the class after that the answers were corrected by me, students and other faculties within 10 minutes. Sample answers were shown in figure 4-7. The answers for the wrong questions were discussed at the end of the class to make them to know the answer of all the questions.

Outcome:

Based on the answers provided by the students, analysis was performed and it is given as a chart in figure 3. From that we can find out that, most of the students had familiarity with the basic concepts of List ADT. Through this activity the students should be able to recall information easily and think individually and improve their problem solving skills.

Relevance Mapping - This activity helps in attaining the following Objectives – PO mapping:

Objectives/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	3	3	-	-	-	-	3	3	-	1



Figure 1



Figure 2

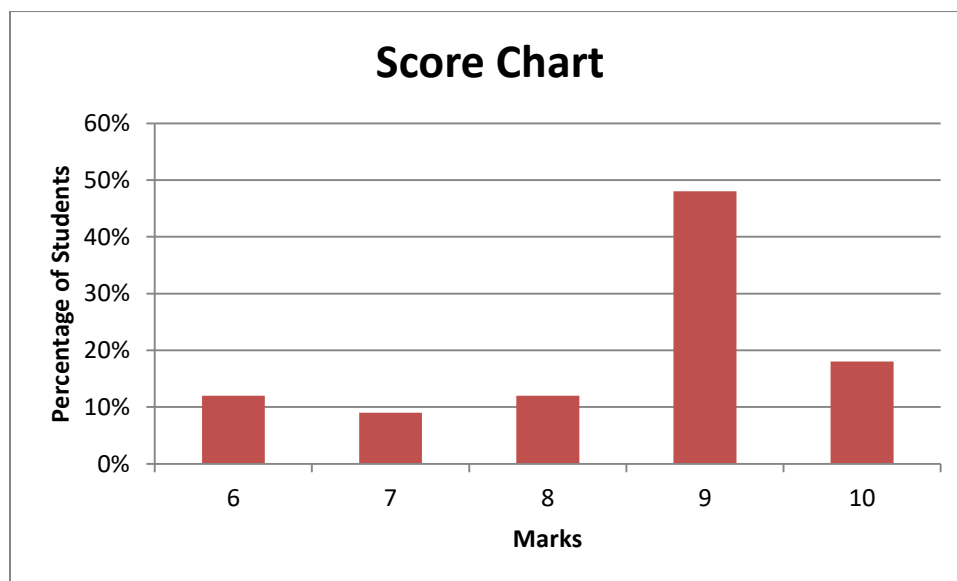
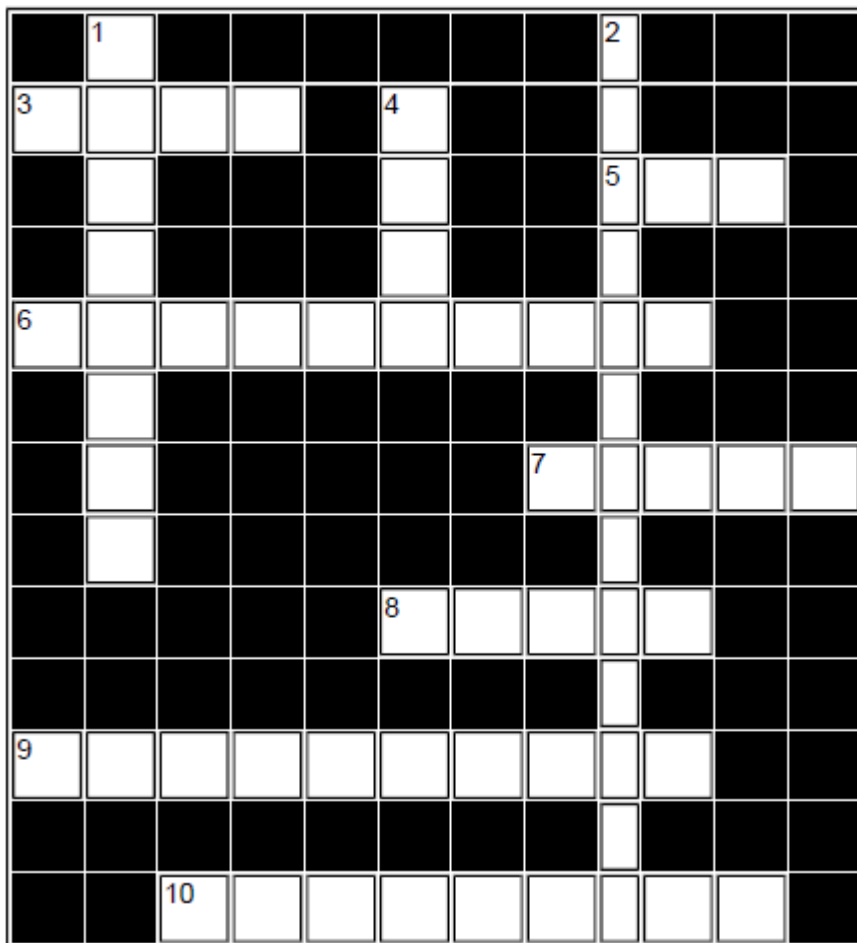


Figure 3: Score sheet for the crossword puzzle

"Linear Data Structure"

Created using the crossword generator at www.classtools.net



ACROSS

3. The Next pointer of last node in a circular linked list contains the address of _____ node. (0,4)
5. Double linked list contains _____ number of pointers in a node. (0,3)
6. Polynomial Operation can be implemented using (0,6,4)
7. What kind of list is best to answer questions such as "What is the item at position n?" (0,5)
8. Example of Linear Data structure (0,5)
9. In a linked list the elements need not be stored in ----- memory. (0,10)
10. Tree Belongs to which type of Data structure (0,3,6)

DOWN

1. _____ node contains address of first node. (0,8)
2. _____ defines a particular way of storing and organizing data in a computer. (0,4,9)
4. What pointer type is used to implement the heterogeneous linked list in C? (0,4)

Answer:

- 1.Sentinel
- 2.Data Structure
- 3.Head
- 4.Void
- 5.Two
- 6.Linked List
- 7.Array
- 8.Stack
- 9.Contiguous
- 10.Non Linear

Sample Student Answers:

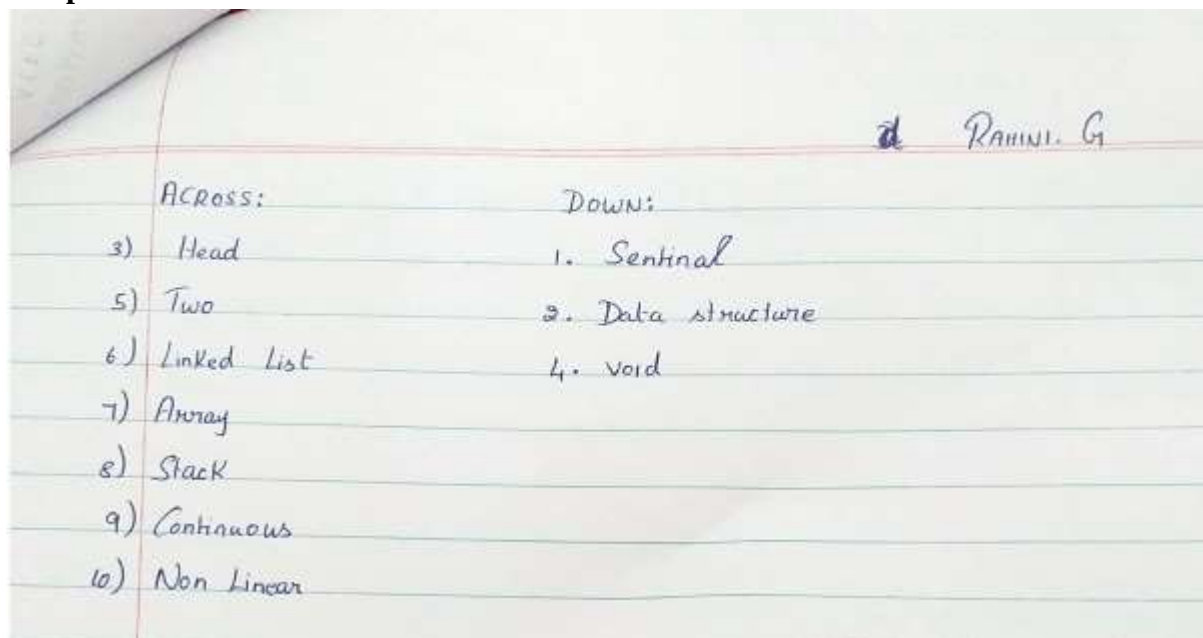


Figure 4: Student answers

CSE - II Y1

3. head
5. two
6. Linked list
7. Array
8. Array / stack / Queue
9. Continuous
10. Non-linear

Down

1. head node
2. Data structure
3. void

1. Previous

2. Data Structure
3. Head
4. void
5. Two
6. Linked List
7. Array
8. Stack
9. Continuous
10. Non-linear

Figure 5: Student answers

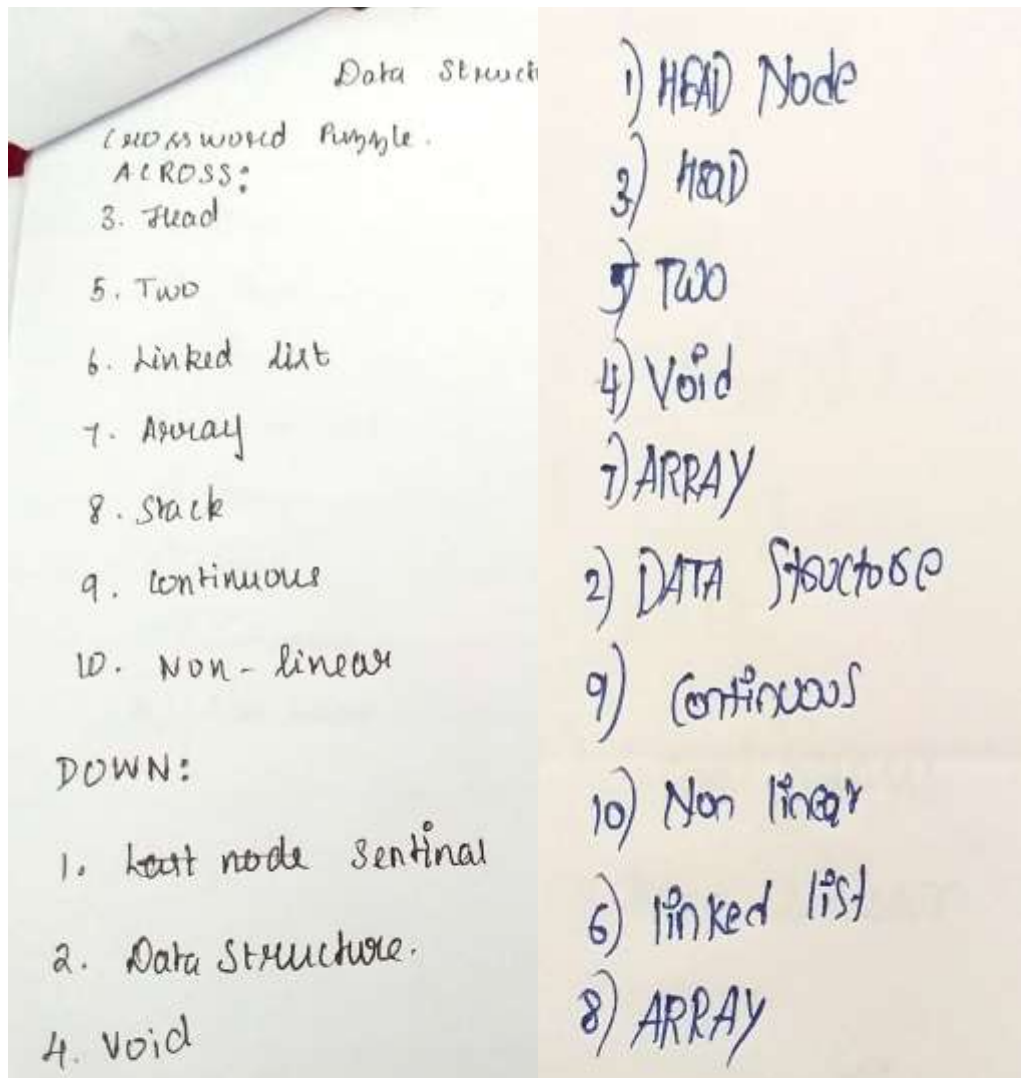


Figure 6: Student answers

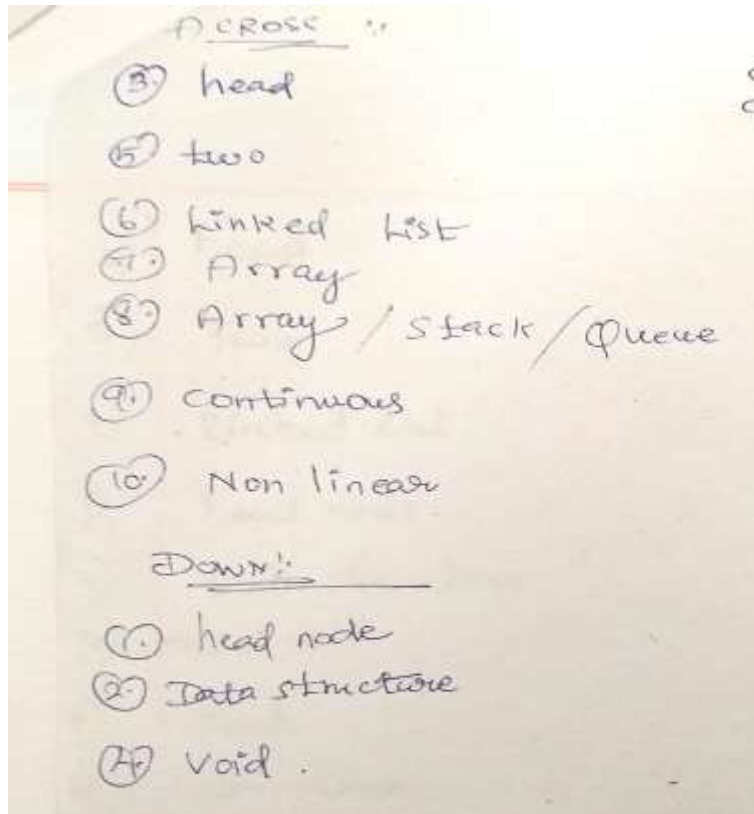


Figure 7: Student answers

Reflective Report

Pre-implementation:

Identified Challenges:

- Students feel difficult by seeing the questions in monitor to match with the crossword section and writing answer in paper.
- Few students do not understand the concept clearly and done a mistake.

Addressing challenges

- I planned to give hardcopy to them individually so that they can write their answers in the crossword itself.

Post implementation

- All the students enjoyed the activity and enthusiastically participated in this activity since it is like a game.
- They can easily remember the basic concepts of linear data structure.

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